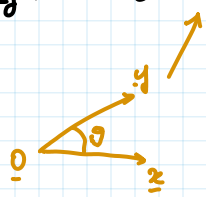
LEZIONE 2

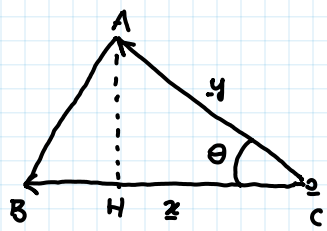
(COSINE SIMILARITY)

Def: Si dice SIMILARITÀ del COSENO la funzione su $\mathbb{R}^N$ data da:

$$d(\underline{x}, \underline{y}) = \cos\theta = \frac{\underline{x} \cdot \underline{y}}{\|\underline{x}\| \cdot \|\underline{y}\|}$$



OSS: Per $N = 2$



$$\overline{AB}^2 = \overline{BC}^2 + \overline{AC}^2 - 2\,\overline{BC} \cdot \overline{AC} \cdot \cos\theta$$

LEGGE del COSENO

$\underline{x} = BC$ $\quad \|\underline{x}\| = \overline{BC}$ $\quad \|\underline{y} - \underline{x}\| = AB$

$\underline{y} = AC$ $\quad \|\underline{y}\| = \overline{AC}$

HW: dimostrare che $d(\underline{x}, \underline{y})$ è data dalla legge del coseno

FUNZIONI in PIÙ VARIABILI

$$\boxed{f : \mathbb{R}^N \longrightarrow \mathbb{R}}$$

$$\underbrace{(x_1, x_2, \dots, x_N)}_{\text{VETTORE di } \mathbb{R}^N} \longmapsto y \in \mathbb{R}$$

$F : \mathbb{R}^N \longrightarrow \mathbb{R}^M$ $\quad F = \begin{bmatrix} f_1 \\ f_2 \\ \dots \\ f_M \end{bmatrix}$ dove $f_i : \mathbb{R}^N \longrightarrow \mathbb{R}$

$$\forall\, (x_1, x_2, \dots, x_N) \in \mathbb{R}^N \ \exists!\, y \in \mathbb{R} : \quad f(x_1, \dots, x_N) = y$$

ESEMPIO per $N = 1$ e $M = 2$

$F : \mathbb{R} \to \mathbb{R}^2$ $\quad F = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$ $\quad f_i : \mathbb{R} \to \mathbb{R}$ $\quad \rho > 0$, $\rho \in \mathbb{R}$

COORDINATE POLARI su cerchio $S(\underline{0}, \rho) = \{ x \in \mathbb{R}^2 : d(\underline{x}, \underline{0}) = \|\underline{x}\| = \rho \}$

$F : [0, 2\pi] \subseteq \mathbb{R} \longrightarrow S(\underline{0}, \rho) \subseteq \mathbb{R}^2$ $\quad F : \theta \longmapsto \underline{x}$ $\quad \underline{x} = \begin{bmatrix} \rho \cos\theta \\ \rho \operatorname{sen}\theta \end{bmatrix}$

ESEMPIO per $N=2$, $M=3$ $\quad F: \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ $\quad F = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}$ $\quad f_i : \mathbb{R}^2 \longrightarrow \mathbb{R}$

COORDINATE SFERICHE $\quad \rho > 0$, $\rho \in \mathbb{R}$

definite sulla SFERA

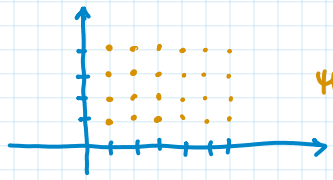
$$S(\underline{0}, \rho) = \{ \underline{x} \in \mathbb{R}^3 : \|\underline{x}\| = \rho \}$$

$$F: [0, 2\pi] \times [0, \pi] \subseteq \mathbb{R}^2 \longrightarrow S(\underline{0}, \rho) \subseteq \mathbb{R}^3$$

$$(\theta, \varphi) \longmapsto \underline{x} \qquad \text{dove } \underline{x} = \begin{bmatrix} \rho \cos\theta \operatorname{sen}\varphi \\ \rho \operatorname{sen}\theta \operatorname{sen}\varphi \\ \rho \cos\varphi \end{bmatrix}$$

HW (Lab): Implementare F per coordinate polari e sferiche con NUMPY e MATPLOTLIB per visualizzare graficamente.

linspace $(0, 100)$ $\rightsquigarrow$ sono le X $\longrightarrow$ calcolare la $f(X) = Y$



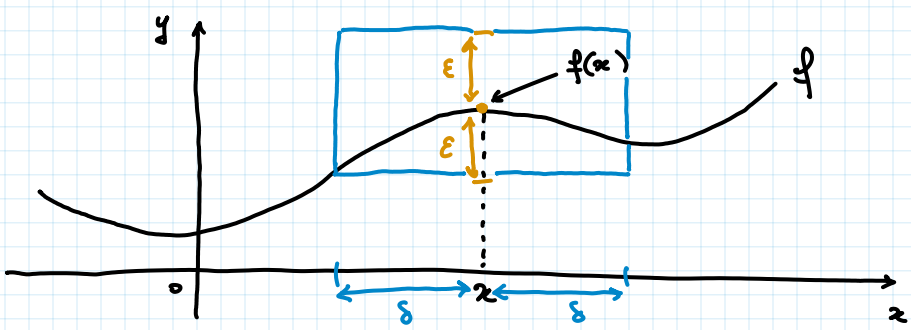
MESHGRID $\longrightarrow$ pti del DOMINIO X, Y $\longrightarrow$ calcolare la $f(X, Y) = Z$

CONTINUITÀ e DERIVABILITÀ

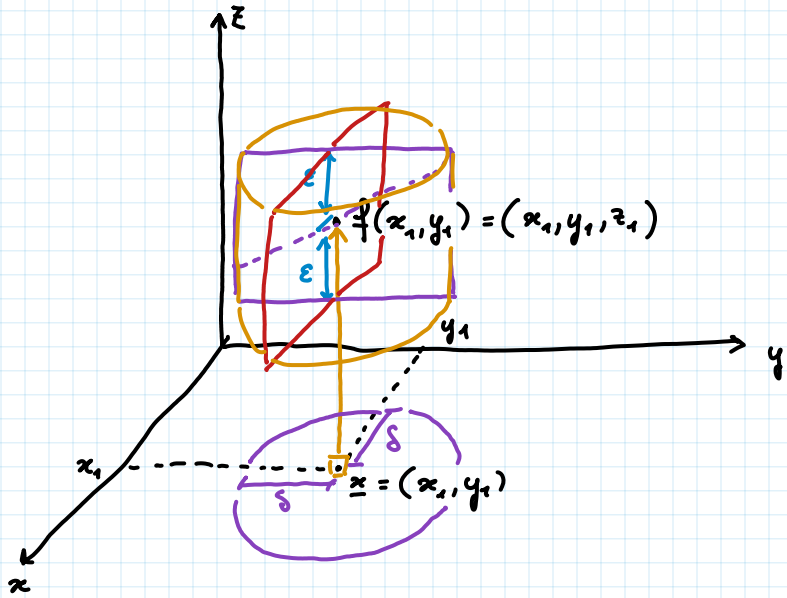
Def: Una funzione $f: \mathbb{R}^N \longrightarrow \mathbb{R}$ si dice CONTINUA su $\mathbb{R}^N$ e si indica $f \in \mathcal{C}^0(\mathbb{R}^N)$, se:

$$\underbrace{\forall \underline{x} \in \mathbb{R}^N}_{\text{dominio di } f} \quad \underbrace{\forall \varepsilon > 0}_{\varepsilon \in \mathbb{R}} \quad \underbrace{\exists \delta > 0}_{\delta \in \mathbb{R}} : \forall \underline{y} \in \mathbb{R}^N, \ \underbrace{\|\underline{x} - \underline{y}\| < \delta}_{d(\underline{x}, \underline{y}) < \delta} \Rightarrow \underbrace{|f(\underline{x}) - f(\underline{y})| < \varepsilon}_{d(f(\underline{x}), f(\underline{y})) < \varepsilon}$$

CASO $N = 1$:



CASO $N=2$:



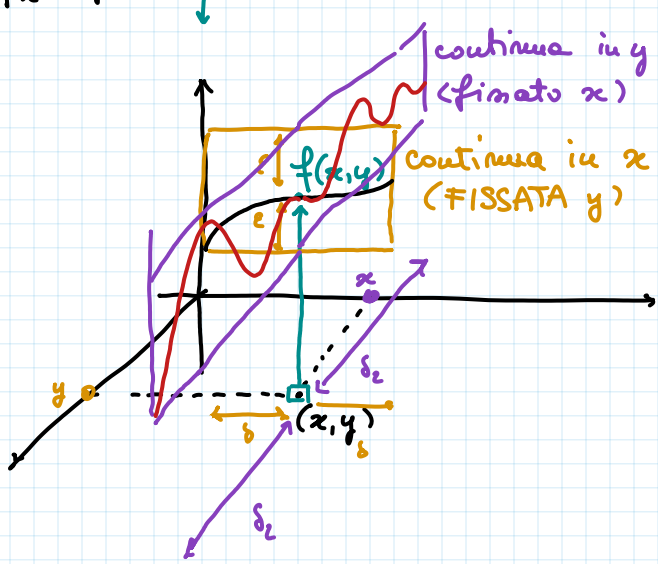
$N=2$

Def: Una funzione $f:\mathbb{R}^2 \to \mathbb{R}$ si dice CONTINUA SEPARATAMENTE in x se, FISSATO y, la funzione

$$f_y = f(\cdot\,, y) : \mathbb{R} \to \mathbb{R} \qquad \text{è CONTINUA su } \mathbb{R}$$

Analogamente si dice CONTINUA SEPARATAMENTE in y se FISSATO x, la funzione

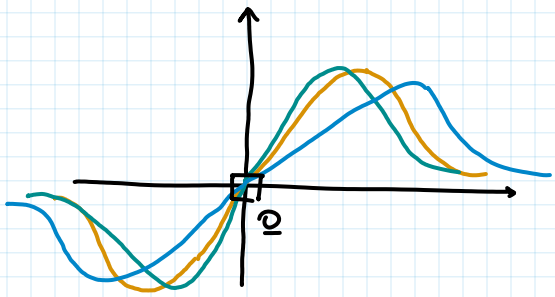
$$f_x = f(x, \cdot\,) : \mathbb{R} \to \mathbb{R} \qquad \text{è CONTINUA su } \mathbb{R}$$



OSS: Ci sono fz continue SEPARATAMENTE in x e y che NON sono CONTINUE su $\mathbb{R}^2$

ESEMPIO: $f : \mathbb{R}^2 \to \mathbb{R}$

$$f(x,y) = \begin{cases} \dfrac{x \cdot y}{x^2+y^2} & \text{se } (x,y) \neq \underline{0} \\ 0 & \text{se } (x,y) = \underline{0} \end{cases}$$



HW: Cosa succede se individuo la direzione (t,t), cioè $x=y$?

Se $x=y$: $\dfrac{x \cdot x}{x^2+x^2} = \dfrac{x^2}{2x^2} = \dfrac{1}{2}$

Def: Una DIREZIONE è un VETTORE $\underline{v} \in \mathbb{R}^N$ t.c. $\|\underline{v}\| = 1$

OGNI vettore di $\mathbb{R}^N$ indica una DIREZIONE:

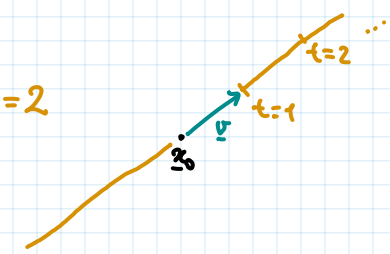
$\underline{x} = (x_1, \dots, x_N)$, $\|\underline{x}\| = k$, prendiamo $\underline{v} = \left(\dfrac{x_1}{k}, \dots, \dfrac{x_N}{k}\right) = \dfrac{\underline{x}}{\|\underline{x}\|}$

allora $\|\underline{v}\| = 1$

Def: Siano $\underline{x}_0 \in \mathbb{R}^N$ e $\underline{v} \in \mathbb{R}^N$, $\|\underline{v}\| = 1$, si dice RETTA passante per $\underline{x}_0$ e direzione $\underline{v}$, l'insieme:

$$R(\underline{x}_0, \underline{v}) = \{ \underline{x} \in \mathbb{R}^N : \underline{x} = \underline{x}_0 + \boxed{t}\,\underline{v}, \ \underline{t \in \mathbb{R}} \}$$

$N=2$



Def: Sia $f : \mathbb{R}^N \to \mathbb{R}$ una funzione, si dice DERIVABILE in $\underline{x}_0 \in \mathbb{R}^N$ e in una direzione $\underline{v} \in \mathbb{R}^N$, $\|\underline{v}\| = 1$, se ESISTE FINITO il <u>limite</u> del rapporto incrementale su $R(\underline{x}_0, \underline{v})$:

$$\frac{\partial f}{\partial \underline{v}}(\underline{x}_0) := \lim_{t \to 0} \frac{f(\underline{x}_0 + t\underline{v}) - f(\underline{x}_0)}{t}$$

DERIVATA DIREZIONALE

OSS: I vettori $\underline{e}_i$ della BASE CANONICA di $\mathbb{R}^N$ sono direzioni

Def: Sia $f:\mathbb{R}^N \to \mathbb{R}$ una funzione, chiamiamo DERIVATA PARZIALE di f rispetto alla i-esima variabile, se esiste, il limite del rapp. incrementale su $R(\underline{x}_0, \underline{e}_i)$:

$$\frac{\partial f}{\partial x_i}(\underline{x}_0) = \lim_{t\to 0} \frac{f(x_1, x_2, \dots, \boxed{x_i + t}, x_{i+1}, \dots, x_N) - f(\underline{x}_0)}{t}$$

$\underline{x}_0 = (x_1, \dots, x_N)$

OSS: Per $N=1$ L'ESISTENZA di f' o $\frac{df}{dx}$ in x_0 garantisce

$\longrightarrow$ f CONTINUA in x_0

$\longrightarrow$ $\exists$ retta tg al GRAFICO di f in x_0

Per $N>1$ L'ESISTENZA di $\frac{\partial f}{\partial x_i}$ per tutte le x_i e nel pto $\underline{x}_0$ da sola NON garantisce

$\not\longrightarrow$ CONTINUITÀ

$\not\longrightarrow$ $\exists$ piano tg al GRAFICO di f

↳ spazio di dim $N-1$

ESEMPIO per $N=2$ $\quad f:\mathbb{R}^2 \longrightarrow \mathbb{R}$ $\quad f(x,y) = x^2 + ye^x$

$\frac{\partial f}{\partial x}(\underline{x}_0) = ?$ $\quad \underline{e}_1 = (1,0)$

$\frac{\partial f}{\partial y}(\underline{x}_0) = ?$ $\quad \underline{e}_2 = (0,1)$

$$\lim_{t\to 0} \frac{f(x+t,y) - f(x,y)}{t} = \lim_{t\to 0} \frac{\overbrace{(x+t)^2}^{\cancel{x^2}+t^2+2xt} + y\overbrace{e^{x+t}}^{e^x e^t} - \cancel{x^2} - ye^x}{t} =$$

$$= \lim_{t\to 0} \frac{t^2 + 2xt + ye^x(e^t - 1)}{t} = \lim_{t\to 0} \frac{\overbrace{t^{\cancel{2}} + 2x\cancel{t}}^{\to 0}}{\cancel{t}} + \lim_{t\to 0} ye^x \underbrace{\boxed{\frac{e^t - 1}{t}}}_{\to 1} = 2x + ye^x$$

$$f(x,y) = x^2 + ye^x$$

$$\frac{\partial f}{\partial y}(\underline{x}_0) = \lim_{t\to 0} \frac{f(x,y+t) - f(x,y)}{t} = \lim_{t\to 0} \frac{\cancel{x^2} + (y+t)e^x - \cancel{x^2} - ye^x}{t} =$$

$$= \lim_{t\to 0} \frac{\cancel{ye^x} + \cancel{t}e^x - \cancel{ye^x}}{\cancel{t}} = e^x$$

$$\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = \left(2x + ye^x, e^x\right)$$

Scegliamo la direzione individuata da $\underline{v}_1 = (1,2) \in \mathbb{R}^2$ ①

$\underline{v}_2 = (3,4) \in \mathbb{R}^2$ ②

$\longrightarrow$ Quanto valgono $\frac{\partial f}{\partial \underline{v}_1}$ e $\frac{\partial f}{\partial \underline{v}_2}$?

① $\underline{v}_1 \rightsquigarrow$ trasformazione $\tilde{\underline{v}}_1 = \frac{\underline{v}_1}{\|\underline{v}_1\|} = \frac{(1,2)}{\sqrt{1^2+2^2}} = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$

$$\frac{\partial f}{\partial \underline{v}_1} = \lim_{t\to 0} \frac{f(x+t, y+2t) - f(x,y)}{t} = \lim_{t\to 0} \frac{(x+t)^2 + (y+2t)e^{x+t} - x^2 - ye^x}{t}$$

$$= \lim_{t\to 0} \frac{\overbrace{t^2}^{\to 0} + \overbrace{2xt}^{2x} + 2\cancel{t}e^{x+t} + \overbrace{ye^xe^t - ye^x}^{\to ye^x}}{\cancel{t}} = \underbrace{\boxed{2x + ye^x}}_{\frac{\partial f}{\partial x}} + 2\underbrace{\boxed{e^x}}_{\frac{\partial f}{\partial y}} = 1 \cdot \frac{\partial f}{\partial x} + 2\frac{\partial f}{\partial y}$$

② $$\frac{\partial f}{\partial \underline{v}_2} = \lim_{t\to 0} \frac{f(x+3t, y+4t) - f(x,y)}{t} \overset{HW}{=} \dots = \underbrace{3}_{\underline{v}_2} \frac{\partial f}{\partial x} + \underbrace{4}_{\underline{v}_2} \frac{\partial f}{\partial y}$$

Quando succede che $\frac{\partial f}{\partial \underline{v}}(x,y) = v_1 \frac{\partial f}{\partial x} + v_2 \frac{\partial f}{\partial y}$ allora f è $\boxed{\text{DIFFERENZIABILE}}$

PER OGNI $\underline{v}$

FUNZIONI DIFFERENZIABILI

Def: Siano $f: \mathbb{R}^N \to \mathbb{R}$ una funzione e $\underline{x}_0 \in \mathbb{R}^N$. Se ESISTONO TUTTE le DERIVATE PARZIALI in $\underline{x}_0$, si può definire il GRADIENTE di f in $\underline{x}_0$, come il VETTORE

$$\nabla f(\underline{x}_0) = \left(\underbrace{\frac{\partial f}{\partial x_1}(\underline{x}_0)}_{\text{direzione } \underline{e}_1}, \underbrace{\frac{\partial f}{\partial x_2}(\underline{x}_0)}_{\underline{e}_2}, \dots, \underbrace{\frac{\partial f}{\partial x_N}(\underline{x}_0)}_{\underline{e}_N} \right)$$

$$\nabla f(\underline{x}_0) \cdot \underline{v} = v_1 \cdot \frac{\partial f}{\partial x_1}(\underline{x}_0) + v_2 \cdot \frac{\partial f}{\partial x_2}(\underline{x}_0) + \dots + v_N \frac{\partial f}{\partial x_N}(\underline{x}_0)$$

Def: Sia $f: \mathbb{R}^N \to \mathbb{R}$ una funzione. SE ESISTE l'applicazione lineare in $\underline{x}_0 \in \mathbb{R}^N$ di direzione $\underline{v}$ detta DIFFERENZIALE

$$\forall \underline{v} \in \mathbb{R}^N: \quad df_{\underline{x}_0}(\underline{v}) \boxed{=} \sum_{i=1}^{N} v_i \frac{\partial f}{\partial x_i}(\underline{x}_0) \boxed{=} \underline{v} \cdot \nabla f(\underline{x}_0) \boxed{= \frac{\partial f}{\partial \underline{v}}(\underline{x}_0)}$$

allora f è DIFFERENZIABILE

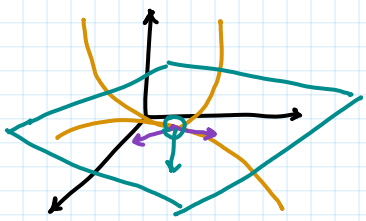
NOTAZIONE: $$df_{\underline{x}_0} = \sum_{i=1}^{N} \frac{\partial f}{\partial x_i}(\underline{x}_0) \cdot dx_i$$

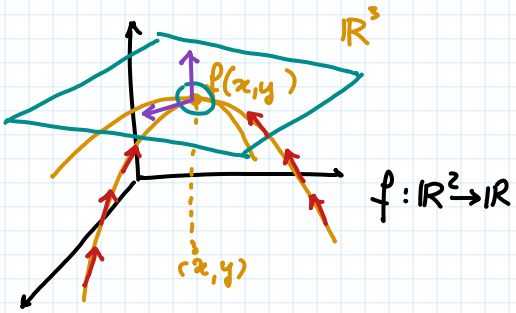
PROPRIETÀ:
- $f: \mathbb{R}^N \to \mathbb{R}$ DIFFERENZIABILE in $\underline{x}_0 \Rightarrow$ CONTINUA in $\underline{x}_0$
- $f: \mathbb{R}^N \to \mathbb{R}$ DIFFERENZIABILE in $\underline{x}_0 \Rightarrow \exists$ il piano tg in $\underline{x}_0$:

$$\varphi(\underline{x}) = f(\underline{x}_0) + \underbrace{df_{\underline{x}_0}}_{\text{differenziale}} \cdot (\underline{x} - \underline{x}_0)$$

OSS: $\nabla f(\underline{x}_0) \cdot \underline{v} = 0$ allora $\nabla f(\underline{x}_0) \perp \underline{v}$

$\hookrightarrow df_{\underline{x}_0} = 0$





PROPRIETÀ: Se $f: \mathbb{R}^N \to \mathbb{R}$ è DIFFERENZIABILE, allora ∇f indica la DIREZIONE di MASSIMA CRESCITA

TEOREMA: Se $f: \mathbb{R}^N \to \mathbb{R}$ è DIFFERENZIABILE e $\underline{x}_0$ è un PUNTO di MASSIMO per f, allora $df_{\underline{x}_0} = 0$ in tutte le dir.
PUNTO di MINIMO

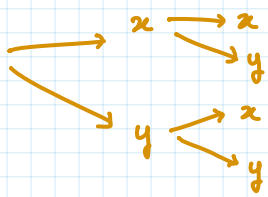
$\hookrightarrow$ Quindi $\nabla f(\underline{x}_0) = 0$

TEOREMA: Sia $f: \mathbb{R}^N \to \mathbb{R}$ con DERIVATE PARZIALI in $B(\underline{x}_0, r)$ con $r > 0$, $r \in \mathbb{R}$, e tali derivate sono CONTINUE
Allora f è DIFFERENZIABILE in $\underline{x}_0$

ESEMPIO: $f: \mathbb{R}^2 \to \mathbb{R}$ $\qquad f(x,y) = x^2 + ye^x$

$$\frac{\partial f}{\partial x} = 2x + ye^x \qquad \frac{\partial f}{\partial y} = e^x$$

DERIVATA SECONDA



$$\frac{\partial^2 f}{\partial x^2} = \text{"}\frac{\partial \partial f}{\partial x \partial x}\text{"} = 2 + ye^x$$

$$\frac{\partial^2 f}{\partial y^2} = 0$$

$$\frac{\partial^2 f}{\partial x \partial y} = e^x = \frac{\partial^2 f}{\partial y \partial x} = e^x$$

TEOREMA di SCHWARTZ: Sia $f: A \subseteq \mathbb{R}^2 \to \mathbb{R}$, con A aperto, e $(x_0, y_0) \in A$, se le derivate MISTE (xy, yx) ESISTONO in $B((x_0,y_0), r)$ e SONO CONTINUE, allora

$$\frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) = \frac{\partial^2 f}{\partial y \partial x}(x_0, y_0)$$